

Griffith School of Engineering and Built Environment
Griffith University

3004ENG — Project Management Principles

Trimester 2, 2026

Eris TestFlight2 Launch Campaign

Project Execution Plan

Assessment 3: Group Project Execution Plan — Group 1

Course	3004ENG Project Management Principles
School	Engineering and Built Environment
Version	Draft
Submission date	2 October 2026
Word count	5,800 (target 6,000 $\pm$ 10%; excludes references, appendices, tables and figures)

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Acting as a project management consultancy engaged to deliver launch-site operations at Bowen Orbital Spaceport.

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1 Executive summary

2 Refined project overview

2.1 Background and purpose

This is the Week 11 reading of a campaign chartered on 7 August 2026. Gilmour Space Technologies attempted the first Australian-made orbital launch with Eris TestFlight 1 from Bowen Orbital Spaceport (Gilmour Space Technologies, 2025; Australian Space Agency, 2025). The vehicle did not reach orbit after a propulsion-side pump failure. TestFlight 2 is the late-2026 follow-on (Dougherty, 2026). Group 1 is engaged to plan and control *launch-site operations*: cleanroom receipt, 23 m stacking, cryogenic GSE, a cold Wet Dress Rehearsal, a five-second hybrid-motor static fire, range clearance, lift-off on 15 November 2026, telemetry handover, and closeout. Manufacturing, engine research, freight to the gate, and pump redesign remain Gilmour factory work (Project Management Institute, 2021; Kerzner, 2017).

The Bowen facility licence has been held since March 2024, so CASA airspace and GBRMPA permissions gate hot fire, not stacking (Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019). The Juru people are High power / High interest, Manage Closely, as the charter already scored them (Project Management Institute, 2021). That class is unchanged. The control is new: monitors with stop-work authority sit on the network before Wet Dress Rehearsal.

2.2 Scope

Scope is the 100% rule on those site operations (Project Management Institute, 2021). Included: campaign governance and EVM (WBS 1.1), pad logistics and GSE (1.2), pad testing (1.3), and flight, launch and closeout (1.4). Seventeen Level-3 packages in Figure 2.1 are the cost accounts; eighteen schedule bars (those seventeen plus external A-100) are the network in Table 3.1. Excluded: factory manufacture, engine R&D, line-haul to the gate, and any unidentified Level-3 box that would mix factory product into a site baseline. A-100 is the boundary object—a zero-dollar wait that drives stacking and is omitted from Table 4.1.

2.3 Charter baseline versus PEP refinement

A refined overview is not a restated charter (Kerzner, 2017). Table 2.1 lists every material change. Two Assessment 1 collisions are closed here: M-4 locked to A-123 on 15 October 2026, and M-8 locked to A-144 on 20 November.

Table 2.1: Charter baseline versus PEP refinement. A1 wording is the Week 4 charter. A3 dates and dollars are the locked YAML values used in Tables 3.1, 4.1 and 6.1.

Project element	A1 charter (Week 4)	A3 PEP (Week 11)	Why it changed
M-4 stacking date	Objective 1: pad integration 15 Oct. Milestone M-4: stacking 28 Oct.	M-4 locked to 15 Oct 2026 (A-123 <i>EF</i>).	Removes the internal collision. Creates a 15-day pad-test buffer before the 30 Oct hot-fire gate.
M-8 handover date	“5 days post-launch” and milestone date 22 Nov (7 days after 15 Nov).	M-8 = 20 Nov 2026 (A-144).	Aligns the milestone with Objective 5 and the 5-day rule.
Cost baseline	“Manage within the allocated operational budget.” No figure.	Bottom-up \$1,500,000 + \$225,000 contingency.	Crane hire, cryogenic crews, and LOX delivery cannot be controlled without a numbered baseline.
Permit vs stacking logic	M-3 flight permits 25 Oct were predecessors of M-4 stacking (then 28 Oct).	Facility licence already held (Mar 2024). A-113 gates A-133 static fire, not stacking. Permits <i>EF</i> 24 Oct; hot fire <i>ES</i> 25 Oct.	Stacking is a site-licence activity. Hot fire is a range activity. This also creates a second critical path.
Juru engagement	High power / high interest, Manage Closely, consultation and monitoring.	Same power/interest (High / High, Manage Closely). Embedded monitors with stop-work authority and a cultural hold point before WDR (A-114 → A-132).	A1 strategy was correct in class; the PEP makes it an enforceable pad control, not a comms plan.
Testing sequence	WDR and 5-second static fire as one Objective 3 output by 30 Oct.	A-132 WDR (cold) 16 Oct–24 Oct, then A-133 static fire 25 Oct–30 Oct. LRR waits on A-134 diagnostics.	Hot fire must wait for flight permits. LRR cannot start on incomplete post-fire data.
Risk governance	Five high-level threats, no $P \times I$ money.	18 campaign-specific risks, Cause→Event→Effect, EMV \$175,500 + RES \$49,500.	Generic rows cannot size contingency or point at a WBS line.
Resource peak	Roles listed; no demand vs capacity.	Organic crew 8 , pad HSE cap 12 , unmitigated peak 14 , mitigated peak 12 .	A1 could not show where the plan exceeds capacity.

Building the network taught the non-date rows. A-113 finishes 24 October and gates static fire, not stacking: that second $TF = 0$ path is why crashing A-123 cannot move 15 November, and locking M-4 to 15 October restores the fifteen-day cold-flow window before the 30 October hot-fire gate. Juru stays Manage Closely, but A-114 now feeds A-132 with stop-work authority. The charter’s five-second static fire remains; it is sequenced after cold WDR rather than fused with it. Cost, risk and occupancy are now numbered: \$1,500,000 plus \$225,000 of EMV-plus-

RES contingency, eighteen cause–event–effect rows, organic crew eight against a pad cap of twelve.

2.4 Objectives and success criteria

Objectives are auditable against the gates in Sections 3–7. Objective 1 is stacking complete on 15 October 2026 (M-4 / A-123). Objective 2 rewrites the Week 4 qualitative budget: *Deliver launch-site operations within the \$1,500,000 AUD base estimate, managing an authorised \$225,000 AUD contingency sized from residual EMV plus a RES uncertainty allowance, audited at Post-Stacking (M-4), Post-Static Fire (M-5), and Post-Launch (M-7)*. Objective 3 separates that 30 October gate: A-132 is cold WDR (16–24 October); A-133 is the five-second static fire (25–30 October). Objective 4 is $T-0$ on 15 November 2026 (A-143 / M-7) with an in-flight slip trigger of more than two days. Objective 5 is telemetry handover on 20 November (A-144 / M-8) and closeout on 1 December (A-145 / M-9).

Finish-line success is a pair of numbers (Project Management Institute, 2021; Kerzner, 2017): final $AC \leq \$1,725,000$ AUD, and $T-0$ on 15 November 2026 with that > 2 day in-flight trigger. Contingency may be consumed; nothing beyond BAC is authorised. A Green CPI of 0.96 in October is allowed under Section 7.5; finishing above \$1,725,000 is not. Those are the same BAC , baseline $T-0$ and two-day trigger used in the earned-value framework.

2.5 Work breakdown structure

The work breakdown structure in Figure 2.1 is the 100% rule applied to launch-site operations at Bowen Orbital Spaceport (Project Management Institute, 2021). Four Level-2 workstreams—campaign governance, pad logistics, pad testing, and flight/closeout—decompose into seventeen Level-3 packages. Every scheduled activity except A-100 maps to exactly one of those packages, and every dollar in Table 4.1 sits in exactly one of them. A-100 is annotated beside the tree rather than inside it: the vehicle receipt window is an external zero-dollar wait that starts when out-of-scope freight reaches the gate. It belongs on the network because it drives stacking (A-122), but it is not a cost account.

Package 1.2.1 is one cost account with two schedule bars (A-121 site mobilisation and A-122 receiving inspection) because receiving cannot start until both the cleanroom and the vehicle exist. The remaining fifteen Level-3 codes are one-to-one with Table 3.1. Factory manufacture, engine research and development, and line-haul to the receiving gate are excluded; parking them as unidentified Level-3 boxes would violate the 100% rule.

Limited notice-to-proceed in the 7 August charter authorised field mobilisation from 10 August. The WBS is therefore live before Gate 1. What Gate 1 (A-111, 2 October) locks is the cost and schedule baseline, not the right to occupy the pad.

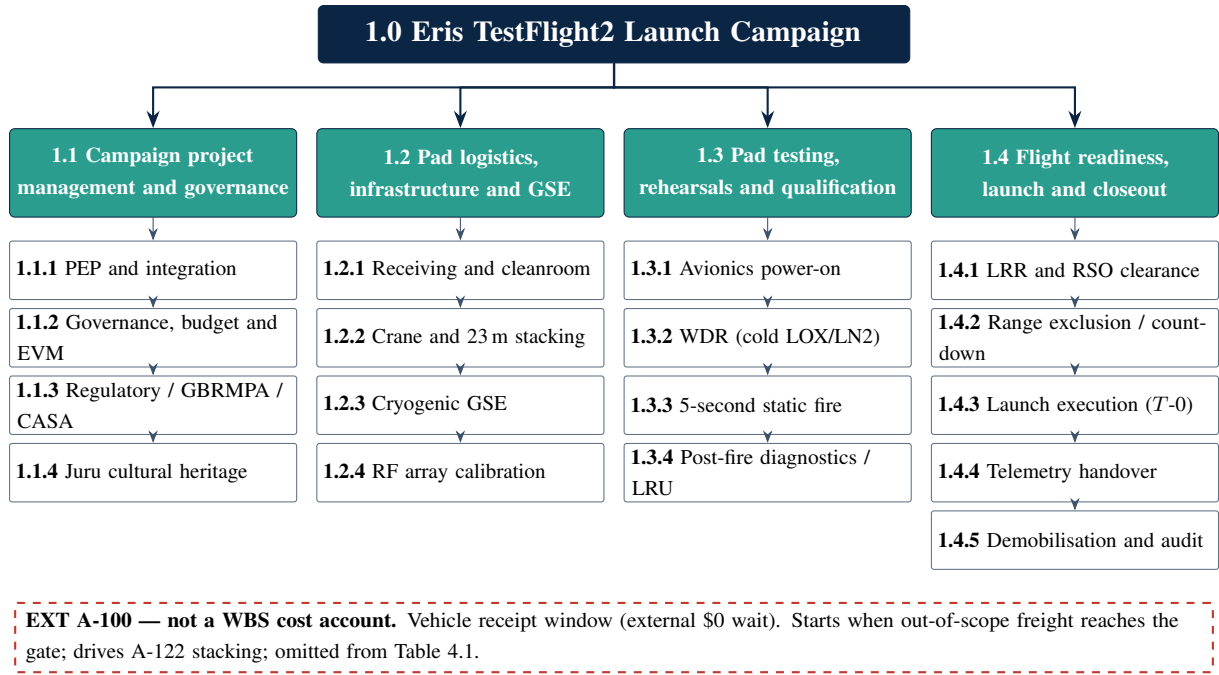


Figure 2.1: Work breakdown structure for Eris TestFlight2 launch-site operations. Seventeen Level-3 packages are the cost accounts in Table 4.1. A-100 is an external schedule bar, not a seventeenth-plus cost line.

3 Schedule management

The campaign schedule converts the WBS into a time-phased baseline for board approval. Dates are derived from a finish-to-start activity-on-node network on a seven-day pad calendar from 10 August to 1 December 2026. Approved $T-0$ remains 15 November 2026. A five-day pull-in to 10 November is a contingent response to payload-window or airspace-slot risk; it is not the baseline.

3.1 Network logic and float analysis

Precedence is modelled as activity-on-node with finish-to-start links and zero lag Project Management Institute (2021); Kerzner (2017). Textbook discrete-time notation writes $EF = ES + D$ when EF is exclusive. On the Bowen calendar every displayed date is inclusive, so the implemented identities are

$$ES_i = \max_{j \in P_i}(EF_j) + 1, \quad EF_i = ES_i + D_i - 1, \quad (1)$$

$$LF_i = \min_{k \in S_i}(LS_k) - 1, \quad LS_i = LF_i - D_i + 1, \quad (2)$$

with P_i and S_i the predecessor and successor sets of activity i , and with ES equal to the campaign start where P_i is empty. Total float and free float then follow as

$$TF_i = LS_i - ES_i = LF_i - EF_i, \quad FF_i = \min_{k \in S_i} (ES_k) - EF_i - 1, \quad (3)$$

taking $FF_i = TF_i$ when S_i is empty. Launch-critical means $TF_i = 0$ on a path that sets $T-0$ or closeout. Table 3.1 is the full pass and matches the locked register.

The forward pass produces two $TF = 0$ chains that remain independent until hot fire. The vehicle path is A-100 (external gate receipt; freight out of scope), then A-122, A-123, A-132 and A-133. The permit path is A-113 into A-133. After A-133 the tail A-134–A-145 is a single critical chain. A-111 finishes on 2 October by an imposed Must-Finish-On (Gate 1) and is not launch-critical: the Week 4 charter already authorised mobilisation.

Path convergence is two staged merges. Wet Dress Rehearsal (A-132) cannot start until stacking (A-123), cryogenic GSE (A-124), avionics power-on (A-131) and Juru monitors (A-114) are complete—the technical merge. Static fire (A-133) then waits on A-132 *and* flight permits (A-113)—the regulatory merge. Launch Readiness Review (A-141) waits on post-fire diagnostics (A-134), not on A-133. Signing a flight-clearance gate on incomplete hot-fire data would break the hold-point logic required of an AS9100D system SAE International (2016).

Feeder float is not interchangeable. A-131 has one day into A-132 and is near-critical; A-124 has three. A-125 (RF) has 39 days and is parked in August–September so it does not sit on the October pad peak; moving it does not change 15 November. A-121 is the instructive case: $TF = 3$ but $FF = 0$, so delay is absorbed before $T-0$ only by consuming A-124's float, while A-124's early start moves immediately. Total float protects $T-0$; free float protects the successor. Figure 3.1 draws the links. Crimson arrows are the $TF = 0$ launch paths; the navy outline on A-111 is Gate 1.

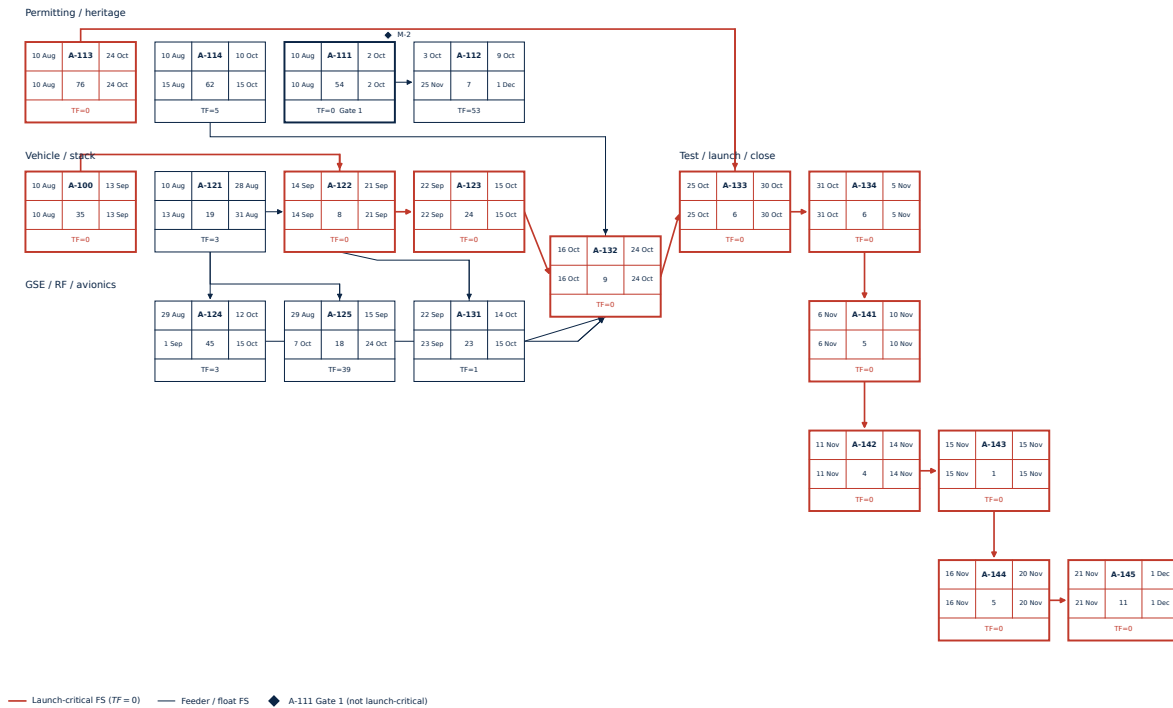


Figure 3.1: Activity-on-node network for the Bowen campaign. Node bands are $ES / ID / EF$, $LS / \text{duration} / LF$, and TF . Crimson arrows are launch-critical finish-to-start links. The vehicle path and the A-113 permit path merge at A-133; A-141 waits on A-134, not on A-133. A-111 is Gate-1 constrained and is not launch-critical.

Table 3.1: Master campaign activity precedence and float register. Dates are inclusive. $FF = \min(ES_{\text{succ}}) - EF - 1$ days, or $FF = TF$ if there is no successor. A-111 $TF = 0$ is an imposed Gate 1 finish, not launch-critical.

ID	WBS	Activity	D	Pred.	ES	EF	LS	LF	TF	FF	CP
A-100	EXT	Vehicle receipt window	35	—	10 Aug	13 Sep	10 Aug	13 Sep	0	0	Y
A-111	1.1.1	PEP and integration baseline	54	—	10 Aug	2 Oct	10 Aug	2 Oct	0*	0	Gate
A-112	1.1.2	EVM control stand-up	7	A-111	3 Oct	9 Oct	25 Nov	1 Dec	53	53	N
A-113	1.1.3	CASA / GBRMPA / airspace	76	—	10 Aug	24 Oct	10 Aug	24 Oct	0	0	Y
A-114	1.1.4	Juru monitor mobilisation	62	—	10 Aug	10 Oct	15 Aug	15 Oct	5	5	N
A-121	1.2.1a	Site mobilisation and cleanroom	19	—	10 Aug	28 Aug	13 Aug	31 Aug	3	0	N
A-122	1.2.1b	Receiving and unpacking	8	A-100, A-121	14 Sep	21 Sep	14 Sep	21 Sep	0	0	Y
A-123	1.2.2	Crane and vertical stacking	24	A-122	22 Sep	15 Oct	22 Sep	15 Oct	0	0	Y
A-124	1.2.3	Cryo GSE piping and checkouts	45	A-121	29 Aug	12 Oct	1 Sep	15 Oct	3	3	N
A-125	1.2.4	RF telemetry calibration	18	A-121	29 Aug	15 Sep	7 Oct	24 Oct	39	39	N
A-131	1.3.1	Avionics power-on sweeps	23	A-122	22 Sep	14 Oct	23 Sep	15 Oct	1	1	N
A-132	1.3.2	Wet Dress Rehearsal (cold)	9	A-114, A-123, A-124, A-131	16 Oct	24 Oct	16 Oct	24 Oct	0	0	Y
A-133	1.3.3	5-second static fire	6	A-132, A-113	25 Oct	30 Oct	25 Oct	30 Oct	0	0	Y
A-134	1.3.4	Post-fire diagnostics and LRU	6	A-133	31 Oct	5 Nov	31 Oct	5 Nov	0	0	Y
A-141	1.4.1	LRR and RSO clearance	5	A-134	6 Nov	10 Nov	6 Nov	10 Nov	0	0	Y
A-142	1.4.2	Range exclusion and countdown	4	A-141	11 Nov	14 Nov	11 Nov	14 Nov	0	0	Y
A-143	1.4.3	Launch execution ($T-0$)	1	A-142	15 Nov	15 Nov	15 Nov	15 Nov	0	0	Y
A-144	1.4.4	Telemetry decrypt and handover	5	A-143	16 Nov	20 Nov	16 Nov	20 Nov	0	0	Y
A-145	1.4.5	Demobilisation and closeout	11	A-144	21 Nov	1 Dec	21 Nov	1 Dec	0	0	Y

3.2 Schedule baseline, Gantt visualisation and milestone integrity

The dates in Table 3.1 *are* the schedule baseline. Figure 3.2 plots the same ES/EF values, colours launch-critical bars crimson, and shows remaining total float as a teal whisker to LF . Permitting, heritage, mobilisation and GSE run through August and September; October is the physical peak (stack 15 October, WDR 16–24 October, static fire 25–30 October); November is the flight-clearance tail.

Two Week 4 collisions are closed here. The charter dated pad integration 15 October in Objective 1 and stacking as M-4 on 28 October; the PEP locks M-4 to A-123 on **15 October 2026**, restoring the cold-flow window before hot fire. Handover “within five days of lift-off” was also dated M-8 at 22 November (seven days after a 15 November $T-0$); M-8 is now A-144 on **20 November**. Because the facility licence is already held, A-113 gates static fire, not stacking. Treating flight permits as a stacking predecessor was an initiation error.

Three gates sit on the baseline. Gate 1 is the PEP lock at A-111 on 2 October. Gate 2 is A-113 on 24 October: without those instruments A-133 cannot start, and that path is already $TF = 0$. Gate 3 is A-141 on 10 November. Crashing stacking without crashing A-113 therefore cannot pull $T-0$.

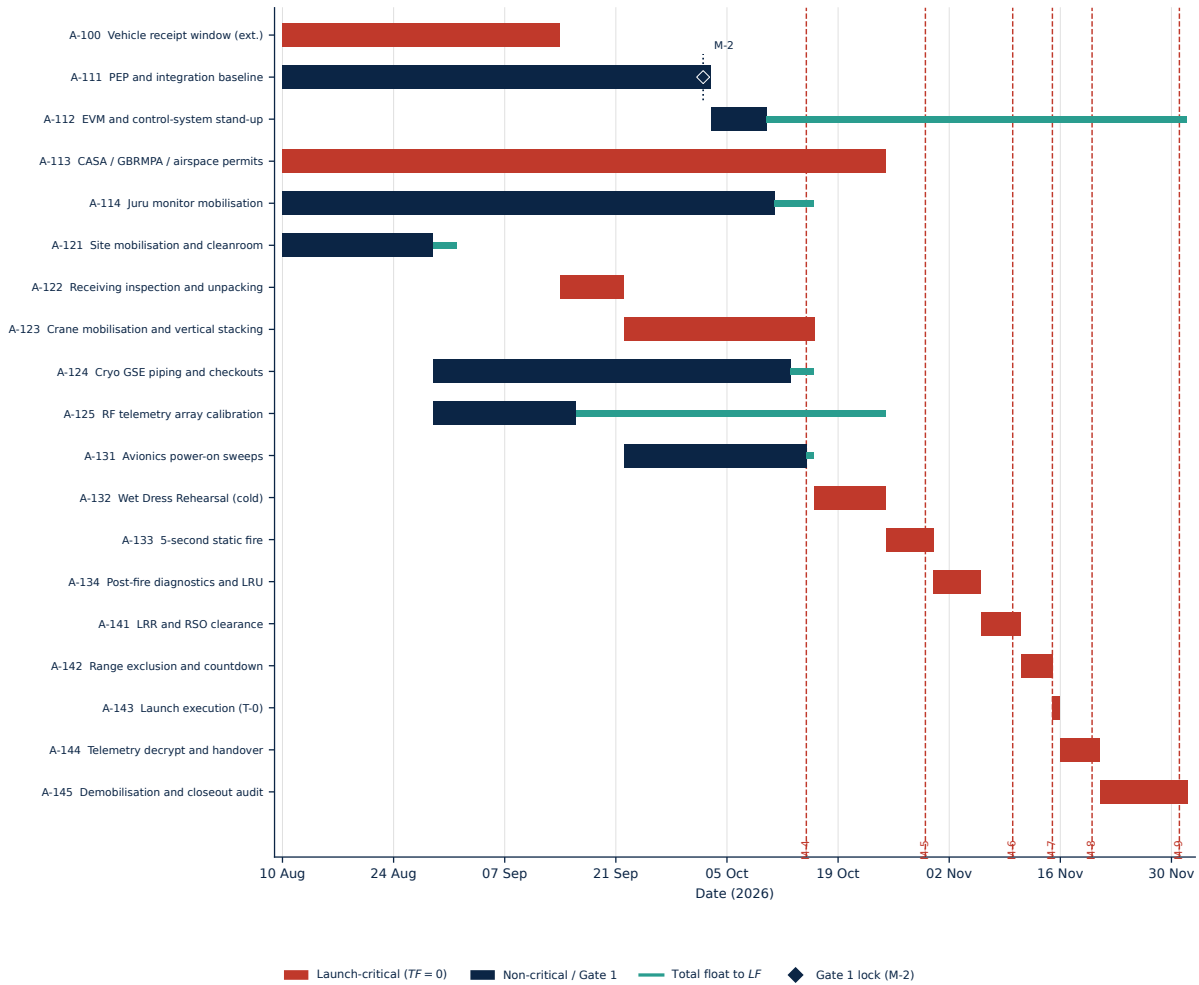


Figure 3.2: Baseline Gantt from 10 August to 1 December 2026, plotted from the same ES/EF values as Table 3.1. Crimson bars are launch-critical. Teal whiskers show total float to LF . Vertical markers are M-2 (navy, Gate 1) and M-4 through M-9. Option C is not drawn on this baseline.

3.3 Schedule compression: crashing and fast-tracking

Kerzner (2017) and Project Management Institute (2021) distinguish fast-tracking (overlap of sequential work, usually at zero direct cost) from crashing (added resource on critical work, paid as a cost slope). The Board question is not weather. It is whether $T-0$ can move from 15 November to 10 November if R-12 is realised (payload LEO geometry valid only 10–12 November) or R-14 is realised (CASA NOTAM refuses the 15 November slot). Five days must come off a $TF = 0$ path that actually sets lift-off. After the A-133 merge that path is the tail A-134–A-143. Crashing A-123 by five days, or overlapping A-123 with A-124, saves **zero** days on $T-0$ while A-113 still finishes on 24 October. A-124 already runs in parallel with stacking.

The crash-cost slope on a crashed activity is

$$\text{Cost slope} = \frac{\Delta C}{\Delta T} = \frac{\text{crash cost} - \text{normal cost}}{\text{normal duration} - \text{crash duration}}. \quad (4)$$

Table 3.2 and Figure 3.3 compare three options that each claim five days. Only Option C is both CPM-valid and acceptable against the LRR hold point.

Table 3.2: Schedule compression evaluation. All three options save five days on $T-0$ if they work as drawn. Baseline $T-0$ remains 15 November; Option C is a contingent draw on the \$225,000 reserve, not an addition to the \$1,500,000 base.

Option	Activities modified	Days	Added cost	Slope	Secondary risks	Verdict
A Pure fast-track	A-134 → A-141 SS+1 (LRR overlaps diagnostics)	5	\$0	\$0/d	LRR before diagnostic close-out; LRU rework; AS9100D hold-point failure	Rejected
B Pure crash	A-134 crashed 6 → 1 d	5	\$40,000	\$8,000/d	Fatigue; skipped thermal/electrical screens; recycle into A-141	Rejected
C Hybrid (if R-12/R-14 fires)	A-134 6 → 3 d (\$9,000); A-141 5 → 3 d (\$12,000); A-142 uncrashed	5	\$21,000	\$4,200/d	Staggered 10-hour shifts; residual diagnostic pressure, buffered by countdown	Contingent

Option A buys five days at zero direct cost by starting A-141 one day after A-134 starts, so the review board would sit while diagnostics were still open. An LRU swap on day four either rubber-stamps incomplete evidence or recalls the review—a hold-point failure under AS9100D SAE International (2016). Option A is rejected.

Option B keeps finish-to-start logic but compresses A-134 from six days to one. The slope is $\$40,000/5 = \$8,000$ per day. A 24-hour diagnostic war room on a vehicle that has just fired is a human-factors problem: thermal and electrical screens are the work most likely to be skipped. The slope is also the dearest of the three. Option B is rejected.

Option C splits the five days across two critical activities after the permit merge: three days off A-134 (\$3,000 per day, \$9,000) and two days off A-141 (\$6,000 per day, \$12,000). The weighted slope is $\$21,000/5 = \$4,200$ per day, about half of Option B. Diagnostics remain three days rather than one; LRR remains a finish-to-start successor; A-142 is not crashed, so the four-day range and countdown buffer before $T-0$ is kept. Fatigue is managed with staggered ten-hour shifts. If invoked, the dates become A-134 31 October–2 November, A-141 3–5 November, A-142 6–9 November, and A-143 on **10 November**. The \$21,000 is drawn from the \$225,000

contingency only if R-12 or R-14 is realised. It is not loaded into the \$1,500,000 base, and Figure 3.2 is not redrawn.

Because Option C does not shorten A-132 or A-133, feeder floats do not tighten: A-131 stays at $TF = 1$, A-124 at $TF = 3$. No secondary critical path is created. Residual exposure is quality pressure on a shorter diagnostic window, which is why A-142 is left intact and why the option remains contingent.

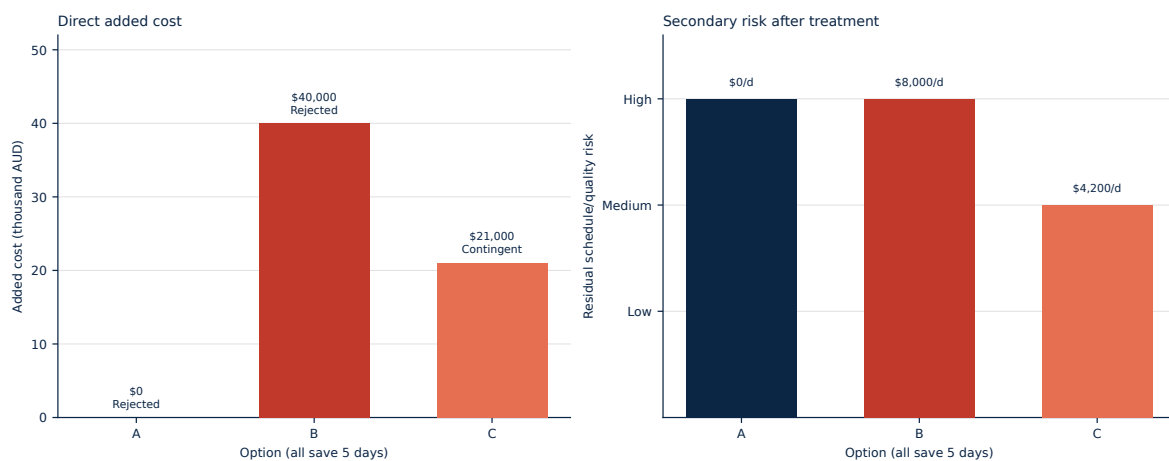


Figure 3.3: Compression trade-off for a five-day pull-in of $T-0$. Left: direct added cost. Right: residual schedule and quality risk after treatment. Option C is the only CPM-valid pull-in that keeps A-142 uncrashed and pays \$4,200 per day rather than \$8,000.

4 Cost management

The cost baseline is the WBS in Figure 2.1 priced at Level 3. Seventeen control accounts sum to a work estimate of \$1,500,000. Contingency of \$225,000 is sized from quantified risk, not from a round-number markup, so the authorised budget at completion is \$1,725,000.

4.1 Estimating basis and assumptions

The estimate is bottom-up at Level 3, with analogous plant and commodity rates beside the hours (Kerzner, 2017; Project Management Institute, 2021). Loaded 2026 Australian-dollar rates are used throughout: systems-engineering and project-management time at \$150/h, pad technicians at \$95/h, cryogenic and range-safety specialists at \$165/h, and surge technicians at \$100/h. Base pay follows the Manufacturing and Associated Industries Award with on-costs for superannuation, leave and workers compensation (Fair Work Ombudsman, 2020). The 100 t crawler for 23 m stacking is priced at \$4,200 per operating day, including operator, from published Queensland plant-hire schedules cross-checked against Rawlinsons (Rawlinsons, 2025).

Inclusions are the seventeen packages in Figure 2.1: mobilisation through demobilisation at Bowen. Factory manufacture, engine research and development, and freight to the receiving

gate remain out of scope, so A-100 is a zero-dollar external wait rather than a cost account. The four-person surge crew (\$22,400) is inside WBS 1.3.3 as planned capacity. The \$21,000 hybrid compression cost in Section 3.3 is a contingent draw against the \$225,000 reserve if R-12 or R-14 is realised; it is not in the \$1,500,000 performance measurement baseline.

A limited notice-to-proceed in the Week 4 charter authorised mobilisation from 10 August. Full cost and schedule lock at Gate 1 (A-111, 2 October). August–September field cost is planned spend, not unauthorised commitment.

4.2 WBS cost baseline

Table 4.1 is the rollup. Pad logistics (1.2) is \$518,000 (34.5% of the work estimate), pad testing (1.3) \$414,000 (27.6%), governance (1.1) \$300,000 (20.0%), and flight/closeout (1.4) \$268,000 (17.9%).

Table 4.1: Bottom-up cost baseline by Level-3 WBS package. Totals equal the work estimate of \$1,500,000. A-100 is omitted because it is not a cost account.

WBS	Package	Amount (AUD)	Basis
1.1.1	PEP and integration	72,000	480 h PM @ \$150
1.1.2	Governance and EVM	48,000	320 h planner/cost @ \$150
1.1.3	CASA / GBRMPA permits	88,000	Consultant + CASA cost-recovery
1.1.4	Juru monitors and heritage	92,000	Two monitors + survey
1.2.1	Receiving and cleanroom	68,000	Technician hours + consumables
1.2.2	Crane and 23 m stacking	218,000	$24 \text{ d} \times \$4,200 = \$100,800$ + riggers
1.2.3	Cryo GSE	180,000	Valves/piping + specialist crew
1.2.4	RF array calibration	52,000	Crew + analyser hire
1.3.1	Avionics power-on	58,000	Engineers @ \$150/h + EGSE
1.3.2	WDR (cold)	160,000	Includes bulk LOX 25 t and LN2 \$68,500
1.3.3	Static fire	148,000	Includes surge hire \$22,400
1.3.4	Post-fire diagnostics	48,000	Engineers + LRU handling
1.4.1	LRR and RSO	38,000	Review board
1.4.2	Range exclusion / marine	58,000	Patrol boats + NOTAM
1.4.3	Launch and telemetry capture	82,000	Launch-day operations
1.4.4	Decrypt and handover	24,000	Five-day engineering handover
1.4.5	Demobilisation and audit	66,000	Site restore + audit
Base		1,500,000	

Notes. Labour: Award base plus on-costs (Fair Work Ombudsman, 2020). Crane and demobilisation: Rawlinsons and Queensland plant-hire (Rawlinsons, 2025). CASA line: agency cost-recovery (Civil Aviation Safety Authority, 2024). Surge $\$22,400 = 4 \times 7 \text{ d} \times 8 \text{ h} \times \100 and sits in 1.3.3, not in contingency.

The largest package is crane mobilisation and stacking (1.2.2, \$218,000) because A-123 occupies 24 calendar days on the launch-critical path: $24 \text{ days} \times \$4,200 = \$100,800$ of hire, with the balance in rigger labour. Wet Dress Rehearsal (1.3.2) carries \$68,500 of bulk liquid

oxygen and nitrogen inside \$160,000. Static fire (1.3.3) carries the planned surge inside \$148,000. Demobilisation (1.4.5) is a Rawlinsons-based site restore plus the closeout audit (Rawlinsons, 2025). Regulatory labour in 1.1.3 includes Civil Aviation Safety Authority cost-recovery (Civil Aviation Safety Authority, 2024). The seventeen lines sum to \$1,500,000 with no miscellaneous remainder. Package 1.2.1 remains one control account for A-121 and A-122.

4.3 Contingency

Contingency is not a 15% story that happens to look quantitative. The 15.00% ratio is the *result* of adding an expected-monetary-value (EMV) pool to a residual uncertainty allowance (Kerzner, 2017). Table 4.2 is the arithmetic. EMV uses the dollar-impact column of the risk register ($EMV = P_{\%} \times I_{\$}$), not the 1–5 ordinal scores. Summing the eleven cost-bearing risks R-01, R-02, R-03, R-04, R-05, R-06, R-07, R-09, R-10, R-13 and R-14 gives \$175,500. An in-scope residual-uncertainty (RES) allowance of \$49,500 covers liquid-oxygen remote-delivery price variance, crane standby in wind, and unquantified pad anomalies. Vehicle freight is out of scope and is not in RES. The two components sum to \$225,000, which is 15.00% of \$1,500,000, and the authorised *BAC* is therefore \$1,725,000.

Table 4.2: Contingency build-up. The 15.00% ratio is the result of EMV plus an in-scope residual allowance, not an input markup.

Component	Amount (AUD)	Rule
Σ EMV of eleven cost-bearing risks	175,500	Dollar EMV from Sec. 6.2; not 1–5 scores
RES uncertainty allowance	49,500	LOX price, crane wind, pad anomalies; not freight
Contingency reserve	225,000	15.00% of base as a <i>result</i>
<i>BAC</i>	1,725,000	Base + contingency

Several register rows are kept out of the \$175,500 so that money is not counted twice. R-08 (pad-civil defect) is contract-transferred under the design-and-construct package and carries no Gilmour EMV. R-11, R-15, R-16, R-17 and R-18 are schedule or HSE exposures without a residual dollar draw. R-12 (payload LEO window) carries a \$21,000 impact equal to Option C; its EMV is held at zero so the compression what-if is not also inside the pre-event total. If R-12 or R-14 is realised and Option C is invoked, the \$21,000 is drawn from the \$225,000 reserve. It is not added to the \$1,500,000 work baseline. There is no management reserve above *BAC*: unknown unknowns would require a board variation this plan does not authorise (Project Management Institute, 2021).

4.4 Time-phased budget and cash flow

The performance measurement baseline is the work estimate spread across the same dates as Table 3.1. Table 4.3 and Figure 4.1 are that spread. August and September (\$218,000 and \$284,000) are charter-authorised mobilisation. October is the peak month at \$657,000 because stacking completes (M-4, 15 October), Wet Dress Rehearsal runs, and the five-second static fire is executed. November carries range, $T-0$ and most of demobilisation (\$315,000). December is a \$26,000 residual on 1 December, not a parking place for unused contingency.

Table 4.3: Time-phased planned value of the \$1,500,000 work baseline. Contingency is not monthly spend.

Month	Period PV (AUD)	Cumulative PV (AUD)
Aug 2026	218,000	218,000
Sep 2026	284,000	502,000
Oct 2026 (peak: stack, WDR, static fire)	657,000	1,159,000
Nov 2026	315,000	1,474,000
Dec 2026	26,000	1,500,000

The S-curve therefore terminates at \$1,500,000. Figure 4.1 draws $BAC = \$1,725,000$ as a horizontal authorisation line. Planned value at the first physical gate (M-4, 15 October) is \$825,000—mid-October, between the September close of \$502,000 and the October close of \$1,159,000. That M-4 figure is the PV used in Section 7.5; it is not a fourteenth weekly data point invented after the fact. Option C's \$21,000 is excluded from every period in Table 4.3. If it is spent, it will appear as actual cost charged to contingency, not as a rewrite of the S-curve.

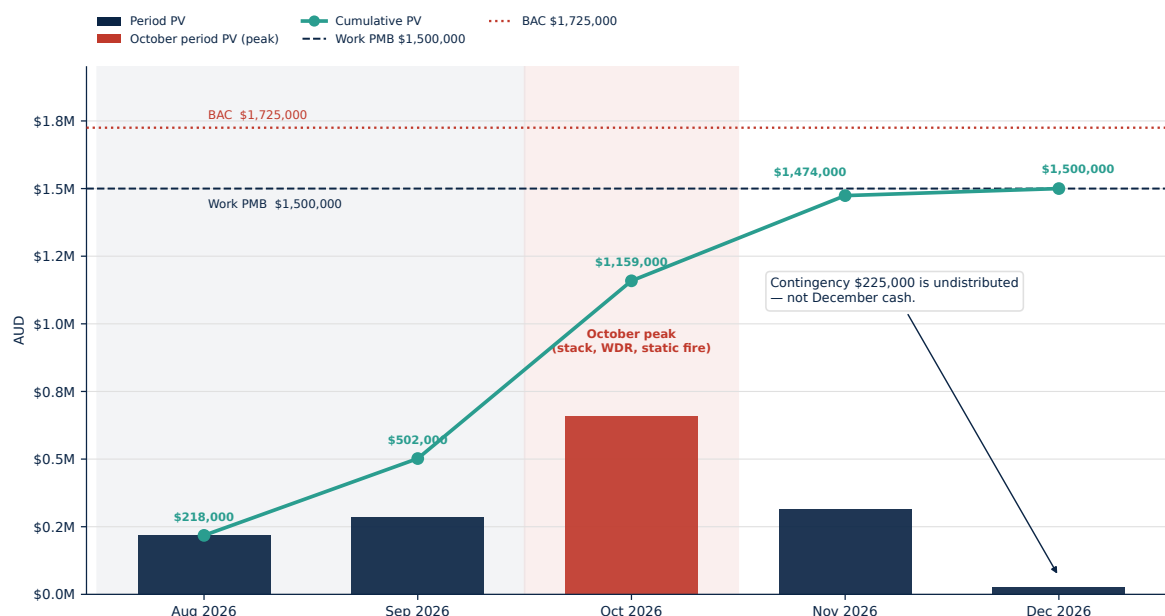


Figure 4.1: Cumulative planned-value S-curve for the Bowen campaign. Period bars are monthly PV; the teal line is cumulative PV and ends at the \$1,500,000 work baseline. The crimson dotted line is *BAC* (\$1,725,000) as an authorisation ceiling, not December cash. Contingency remains undistributed. Option C is not in the curve.

5 Resource management

Pad occupancy is a health-and-safety limit, not a roster convenience. The organic Bowen crew is eight certified technicians. The blast-zone, suit-air and access-control cap is twelve people. Eight is therefore the standing organisation; twelve is the occupancy ceiling that Figure 5.1 must never cross (Project Management Institute, 2021; Kerzner, 2017).

5.1 Roles and responsibilities

Accountability is split across two layers, with exactly one Accountable owner per Level-3 package in Table 5.1. The consultancy layer is Abdul Wahab Obeid (campaign PM and integration), Hadi Alkhub (planner and schedule), Ziyad Alahmari (risk, quality and HSE) and Ilsa Hashmi (cost, resources and the Juru commercial interface). The field layer is the Launch Director, Range Safety Officer, GSE Lead, Avionics Specialist, Juru Cultural Officer and the design-and-construct package manager. Consulted names are the people who must see the work product before the Accountable owner signs; they are not a second A.

The pattern matches the network. Governance packages sit with the consultancy: Abdul on the PEP and on CASA/GBRMPA permits, Ilsa on EVM, the Juru Cultural Officer on heritage with Ilsa consulted. Pad logistics sit with the people who physically control the lift and the interfaces: Hadi is Accountable for stacking because the dawn-lift window is a schedule control (R-06); GSE and D&C share cryogenic hardware; avionics owns RF and power-on. Hot operations

(static fire, LRR, $T-0$) are Accountable to the Launch Director with the RSO consulted, except range exclusion itself, which the RSO owns. Closeout audit is Ziyad's, so quality is not marking its own homework against a field lead.

Table 5.1: RACI for Level-3 packages. Exactly one Accountable owner per row. Field titles are as used on the pad; consultancy names are the Group 1 officers.

WBS	Package	A	R	C
1.1.1	PEP and integration	Abdul	Hadi	Ziyad, Ilsa
1.1.2	Governance and EVM	Ilsa	Hadi	Abdul
1.1.3	CASA / GBRMPA permits	Abdul	Hadi	Launch Director
1.1.4	Juru heritage	Juru Officer	Ilsa	Abdul, Ziyad
1.2.1	Receiving / cleanroom	GSE Lead	D&C mgr	Hadi, Ilsa
1.2.2	Crane and stacking	Hadi	GSE Lead	Launch Director, RSO
1.2.3	Cryo GSE	D&C mgr	GSE Lead	Ziyad
1.2.4	RF calibration	Avionics	Hadi	Launch Director
1.3.1	Avionics power-on	Avionics	GSE Lead	Hadi
1.3.2	WDR (cold)	GSE Lead	Juru Officer	Ziyad, LD, RSO
1.3.3	Static fire	Launch Director	GSE Lead	RSO, Ziyad, Avionics
1.3.4	Post-fire / LRU	Avionics	GSE Lead	Ziyad
1.4.1	LRR and RSO	Launch Director	RSO	Abdul, Ziyad
1.4.2	Range exclusion	RSO	Launch Director	Juru Officer
1.4.3	Launch ($T-0$)	Launch Director	RSO, Avionics	Abdul
1.4.4	Telemetry handover	Avionics	Hadi	Ilsa
1.4.5	Demob and audit	Ziyad	D&C mgr	Ilsa, Abdul

5.2 Capacity planning

Figure 5.1 is the weekly demand implied by Table 3.1 against those two headcount numbers. If RF calibration (A-125) remained stacked on Wet Dress Rehearsal and static fire, weeks 10–12 would peak at fourteen people: two above the HSE cap and six above the organic crew. That is R-13 in numeric form, not a generic “over-allocation.”

The baseline does two things, in that order. First, A-125 is placed in weeks 3–6 (29 August to 15 September) using its 39 days of total float. That move costs \$0 and does not change 15 November, because A-125 is not on the launch-critical path. Second, four surge technicians are hired for seven days inside WBS 1.3.3 at \$100/h loaded: $4 \times 7 \times 8 \times \$100 = \$22,400$, already inside the \$1,500,000 work estimate (Fair Work Ombudsman, 2020). The mitigated series then peaks at twelve in weeks 11–12, on the cap and never above it. Organic eight remains the standing crew; the extra four are a planned pulse, not evidence that the cap was quietly raised. Staggered ten-hour shifts in those two weeks keep twelve people from occupying the blast zone at once even though the daily roster is twelve. Option C, if invoked, shortens A-134 and A-141 and does not redraw Figure 5.1: the surge technicians remain a 1.3.3 base cost, not a compression

cost. Moving A-121 or A-124 would consume feeder float without lowering the week-11 peak, so they are not used as capacity tools.



Figure 5.1: Weekly pad demand versus capacity. Hashed crimson bars are the unmitigated series (peak 14 in weeks 10–12). Teal bars are the mitigated baseline (peak 12) after moving A-125 into weeks 3–6 and adding four surge technicians in 1.3.3. The navy dashed line is the organic crew of eight; the crimson dotted line is the HSE occupancy cap of twelve. Eight is not a hard pad limit.

5.3 Conflict resolution and fallback

Resource conflicts escalate on the work, not on personalities. Tier 1 field leads have four hours to level, stagger or borrow from float without spending money. Tier 2 is the campaign PM: twelve hours to choose overtime, hire or resequence, with the cost consequence going to Ilsa so that Table 4.1 stays the book of record. Tier 3 is the Executive Board, with a 24-hour SLA, for any decision that consumes contingency or moves M-5 (30 October) or M-7 (15 November). Those gates are the same dates as Section 3; they are not a second calendar.

Team-member fallback is separate and is the A1 defect closed here. If a member is non-responsive for more than 48 hours, the Lead Planner audits the artefacts within 12 hours, the PM issues a WBS-reassignment addendum, and the Buddycheck PAF is locked against Teams and version history so that contribution evidence cannot be rewritten after the fact. That procedure reallocates packages in Table 5.1; it does not authorise a silent change to *T-0* or to *BAC*.

6 Risk management

Campaign risk is owned by Ziyad Alahmari (Risk and Quality Lead) and is scored on this pad, not as generic weather or budget-overrun rows. Eighteen cause–event–effect statements

in Table 6.1 size the \$225,000 contingency already reported in Table 4.2, name the float that absorbs non-critical delay, and fix the owners who treat each item.

6.1 Risk approach and scales

The process follows ISO 31000 identify–analyse–evaluate–treat–monitor (International Organization for Standardization, 2018). Expected monetary value is calculated as in Kerzner (2017): money is never read from the 1–5 heat-map scores. $EMV = P_{\%} \times I_{\$}$ uses the dollar column in Table 6.1. Probability bands are $P = 1$ below 5% (3% in EMV), $P = 2$ from 5–20% (12%), $P = 3$ from 21–40% (30%), $P = 4$ from 41–60% (50%), and $P = 5$ above 60% (70%). Schedule impact I_S is 1 (< 2 d), 2 (2–5 d), 3 (6–10 d), 4 (11–15 d) and 5 (> 15 d on a $TF = 0$ path). Cost impact I_C is the dollar figure in the same row. The matrix coordinate I is the maximum of those two bands (Project Management Institute, 2021). Residual P and I fall only where the named response actually bites. R-12 keeps $P = 2$ because Option C does not make the payload window more likely; it only prices the pull-in. Ziyad re-scores the register at Gate 1 (2 October), at M-4 (15 October) and at M-5 (30 October), in the same cadence as the EVM dashboard in Section 7.5.

6.2 Integrated risk register

Table 6.1 is the full set. Cause, event and effect are three clauses. Owners match the RACI in Table 5.1: Ziyad with the GSE Lead on static-fire abort (R-01) and township acoustics (R-17); the Launch Director on footprint and stage-separation (R-02, R-09); Ilsa Hashmi on LOX logistics, occupancy and the Juru commercial interface (R-10, R-13, R-04 with the Juru Cultural Officer); Hadi Alkhub on the dawn-lift window and countdown software (R-06, R-18); Abdul Wahab Obeid on permits, LRR deadlock and the Option C draw (R-14, R-16, R-12). Figure 6.1 plots every inherent point and the residual after treatment. R-02 and R-09 share $(P, I) = (2, 5)$ and are jittered so both remain visible.

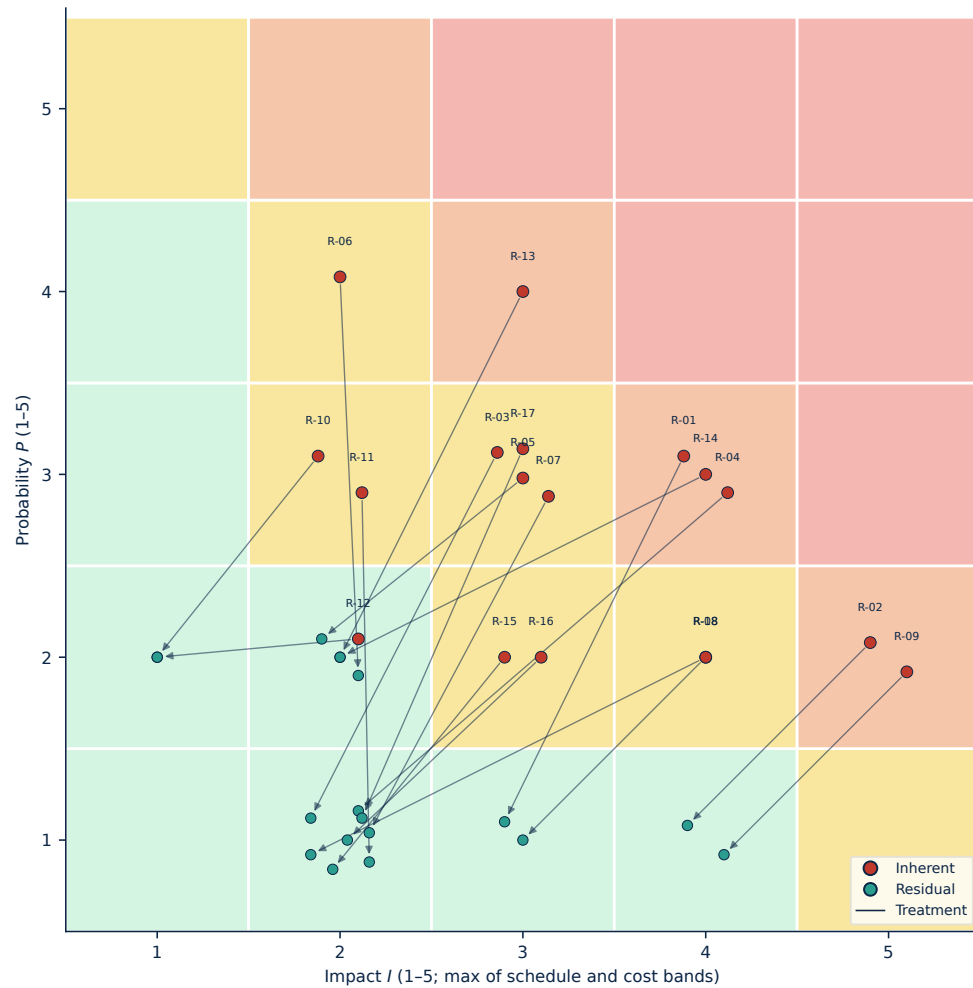


Figure 6.1: Inherent (crimson) to residual (teal) movement for all eighteen register items. P is vertical and I is horizontal. Most treatment arrows run down and left. R-12 keeps $P = 2$ (the payload window is not made less likely) and moves I only. R-02 and R-09 share the (2, 5) cell and are offset.

Table 6.1: Integrated campaign risk register. $P \times I$ uses the 1–5 scores. EMV uses $P_{\%} \times I_{\$}$, not those scores. A dash means the row is not in the \$175,500 pool.

ID	Anchor	Cause → event → effect	P	I	$P \times I$	Owner	Response	P'	I'	EMV
R-01	A-133 / 1.3.3	Residual pump insulation → static-fire auto-abort → 12–14 d recycle / \$90k	3	4	12	Ziyad / GSE Lead	Thermal soak; on-pad LRU kit; 48 h recycle procedure	1	3	27,000
R-02	A-143 / 1.4.3	Trajectory energy toward park boundary → debris in GBRMPA zone → statutory halt	2	5	10	Launch Director	South-biased LRR footprint; joint GBRMPA protocol	1	4	21,600
R-03	A-125 / A-143	RF interference at Bowen array → telemetry <98.5% → heritage fail	3	3	9	Avionics Lead	Dual-path recorders; A-125 before hot fire; mobile backup	1	2	15,000
R-04	A-114 / A-143	Exclusion without stop-work → Juru complaint → range withheld	3	4	12	Ilsa / Juru Officer	Embedded monitors with stop-work; cultural HP before A-132	1	2	12,000
R-05	A-124 / A-132	Inadequate purge before LOX → frozen/cracked valve → WDR abort ~8 d	3	3	9	GSE Lead	LN2 pre-chill; spare valve kit; helium leak test HP-2	2	2	16,500
R-06	A-123 / 1.2.2	Abbot Point crosswind >20 kt at vertical mate → lift abort → crane standby	4	2	8	Hadi / Stack Conductor	Dawn-lift window; standby already inside 1.2.2	2	2	9,000
R-07	A-124 / D&C	D&C GSE IFC lags propulsion ICD → pad mismatch → rework	3	3	9	GSE Lead / PM	ICD freeze gate; weekly integration; D&C LDs	1	2	21,000
R-08	1.2.3 / D&C	Latent pad-civil defect → hold-down failure → hot-ops stop	2	4	8	D&C mgr (A: PM)	NDT HP-1; defect liability remains with D&C	1	2	0*

Table 6.1: Integrated campaign risk register (continued).

ID	Anchor	Cause → event → effect	<i>P</i>	<i>I</i>	<i>P</i> × <i>I</i>	Owner	Response	<i>P'</i>	<i>I'</i>	EMV
R-09	A-143	Stage-sep charge timing error → debris outside footprint → inquiry / halt	2	5	10	Launch Director	Dual-string sep inhibit; wind hold toward park	1	4	18,000
R-10	A-132 / 1.3.2	Single 800 km LOX tanker delay → WDR cannot load → 2–5 d slip	3	2	6	Ilsa	72 h tanker confirm; Townsville backup; RES on price	2	1	6,900
R-11	A-131	Flight-computer vs pad EGSE skew → power-on abort	3	2	6	Avionics Specialist	ICD freeze; cleanroom EGSE dry run before stack	1	2	—
R-12	A-143	Payload LEO geometry valid only 10–12 Nov → board directs 5-day pull-in	2	2	4	Abdul / Sponsor	Pre-priced Option C; draw \$21k from contingency if approved	2	1	—
R-13	A-132 / A-133	Peak 14 vs cap 12 → detank error → recycle / injury	4	3	12	Ilsa	A-125 moved to Sep; four surge techs in base; 10 h stagger	2	2	15,000
R-14	A-113 / A-143	Townsville airspace / late NOTAM → 15 Nov slot refused	3	4	12	Abdul / Regulatory	File 60 d prior; weekly CASA liaison; Option C backup	2	2	13,500
R-15	A-142	Dugong in exclusion box → sweep extension	2	3	6	RSO	Marine-mammal protocol; A-142 holds 4 d vs <i>T</i> -0	1	2	—
R-16	A-141	RSO vs Launch Director deadlock at LRR → missed 10 Nov gate	2	3	6	Abdul / Launch Director	Named DoA; escalate 24 h before A-141 <i>LF</i>	1	2	—

Table 6.1: Integrated campaign risk register (continued).

ID	Anchor	Cause → event → effect	<i>P</i>	<i>I</i>	<i>P</i> × <i>I</i>	Owner	Response	<i>P'</i>	<i>I'</i>	EMV
R-17	A-133	Static-fire acoustic >135 dB at Bowen → complaint / hold	3	3	9	Ziyad / HSE	Water deluge; 1.5 km exclusion; community notice 14 d prior	1	2	—
R-18	A-142 / A-143	FTS/countdown abort at <i>T</i> -10 → recycle inside window	2	4	8	Hadi / Avionics	Two countdown rehearsals; abort-to-recycle procedure	1	3	—

*R-08 is contract-transferred (EMV \$0 to Gilmour). R-11, R-12, R-15–R-18 are schedule/HSE or priced only if Option C is invoked. EMV identities: R-01 $0.30 \times 90,000$; R-02 $0.12 \times 180,000$; R-03 $0.30 \times 50,000$; R-04 $0.30 \times 40,000$; R-05 $0.30 \times 55,000$; R-06 $0.50 \times 18,000$; R-07 $0.30 \times 70,000$; R-09 $0.12 \times 150,000$; R-10 $0.30 \times 23,000$; R-13 $0.50 \times 30,000$; R-14 $0.30 \times 45,000$. Sum \$175,500.

6.3 Top risks, contingency, float and delivery model

Three rows dominate board attention because they sit on the launch-critical path and can stop $T-0$ or close the range. They are not a recycled cyclone or “budget overrun” list. R-01 is a residual pump-insulation defect on A-133: auto-abort, a 12–14 day recycle, about \$90k of consumables. Ziyad and the GSE Lead treat it with thermal-soak instrumentation, an on-pad LRU kit and a 48-hour recycle procedure, which drops the score from (3, 4) to (1, 3) and leaves \$27,000 in the EMV pool. R-09 is a stage-separation timing error at A-143: debris outside the predicted footprint, GBRMPA inquiry, range halt. The Launch Director owns a dual-string inhibit and a wind hold toward the park; residual remains high-impact (1, 4) because a statutory halt does not become cheaper when the probability falls. R-02 is the sibling footprint risk (trajectory energy toward the park boundary) and is treated on the same LRR protocol; both remain Extreme-to-High on impact. R-14 is Townsville military airspace or a late NOTAM refusing the 15 November slot. Abdul files sixty days prior and holds a weekly CASA liaison. Crashing A-123 does not treat R-14: permits already finish on 24 October with $TF = 0$, so a faster stack still waits on A-113. Option C is the schedule backup, not a rewrite of the baseline.

Those eleven cost-bearing rows sum to \$175,500. Ilsa’s RES allowance of \$49,500 covers in-scope LOX price variance, crane wind standby and unquantified pad anomalies. Together they are the \$225,000 in Table 4.2. Vehicle freight stays out of RES because it is out of scope. R-06 (Abbot Point crosswind at vertical mate) is a named lift-window risk owned by Hadi; its standby dollars already sit inside WBS 1.2.2, which is why the EMV is \$9,000 rather than a second crane line.

Float, not money, is the treatment for several feeders. A-125’s 39 days let Ilsa park RF in weeks 3–6 so R-13 never meets an occupancy of 14. A-124’s three days and A-114’s five days absorb GSE and heritage delay before A-132 without moving 15 November. R-08 is transferred: the D&C package manager remains Responsible, Abdul remains Accountable, and defect liability stays with the contractor after HP-1 NDT. R-12 is accepted on probability and pre-priced on cost. If the payload window is only 10–12 November, or if R-14 fires, Abdul takes Option C to the Board and Ilsa draws \$21,000 from contingency. That draw is not in the \$1,500,000 work baseline, and it does not move the M-7 diamond on Figure 3.2 until the Board says so.

7 Delivery and control

This section converts the baselines in Sections 3–6 into named decision rights, a procurement choice that fits this pad, verification hold points, Bowen-specific HSE, and an earned-value framework that will be read at M-4. It is a control design, not a progress report: the campaign has not been executed.

7.1 Governance and reporting

Governance is the organisation that will actually stop work, not a generic pyramid (Project Management Institute, 2021; Kerzner, 2017). The Gilmour Executive Board is the sponsor. Abdul Wahab Obeid is Campaign Project Manager: he reports to the Board and directs three consultancy leads already named in Table 5.1—Hadi Alkhub (planner and schedule), Ziyad Alahmari (risk, quality and HSE) and Ilsa Hashmi (cost, resources and the Juru commercial interface). Beneath them the pad is run by the field titles used throughout the register: Launch Director, Range Safety Officer, GSE Lead, Avionics Specialist, Juru Cultural Officer and the design-and-construct package manager. Figure 7.1 is that structure. The DoA sits on the PM box because that is the only place a dollar-and-hour rule can be enforced.

Delegation is numeric. Abdul may approve a variance of at most \$20,000 *and* at most 48 hours, and only if M-5 (30 October, A-133) and M-7 (15 November, A-143) do not move. Anything else—a contingency draw, Option C's \$21,000, a hold-point waiver, a *T-0* change—goes to the Board within 24 hours (Kerzner, 2017). That is also the treatment of R-16: an RSO versus Launch Director deadlock at LRR escalates 24 hours before A-141's late finish of 10 Nov, not inside the review room. The \$20,000 ceiling is below the dollar *impact* of the hot-fire rows that would recycle the pad (R-01 \$90,000; R-07 \$70,000). Those rows' EMVs are \$27,000 and \$21,000; a genuine recycle cannot be buried in PM discretion.

Cadence matches the physical work. Field leads hold a daily pad standup (yesterday, today, open holds). After Gate 1 (A-111, 2 Oct) Ilsa publishes a weekly EVM dashboard—CPI, SPI, *EAC*, hold-point status—from A-112. Abdul takes a fortnightly pack to the sponsor. Daily standup does not rewrite Table 3.1; the weekly dashboard is where Amber is first declared. A-112 has 53 days of float so the control system does not compete with stacking for pad occupancy.

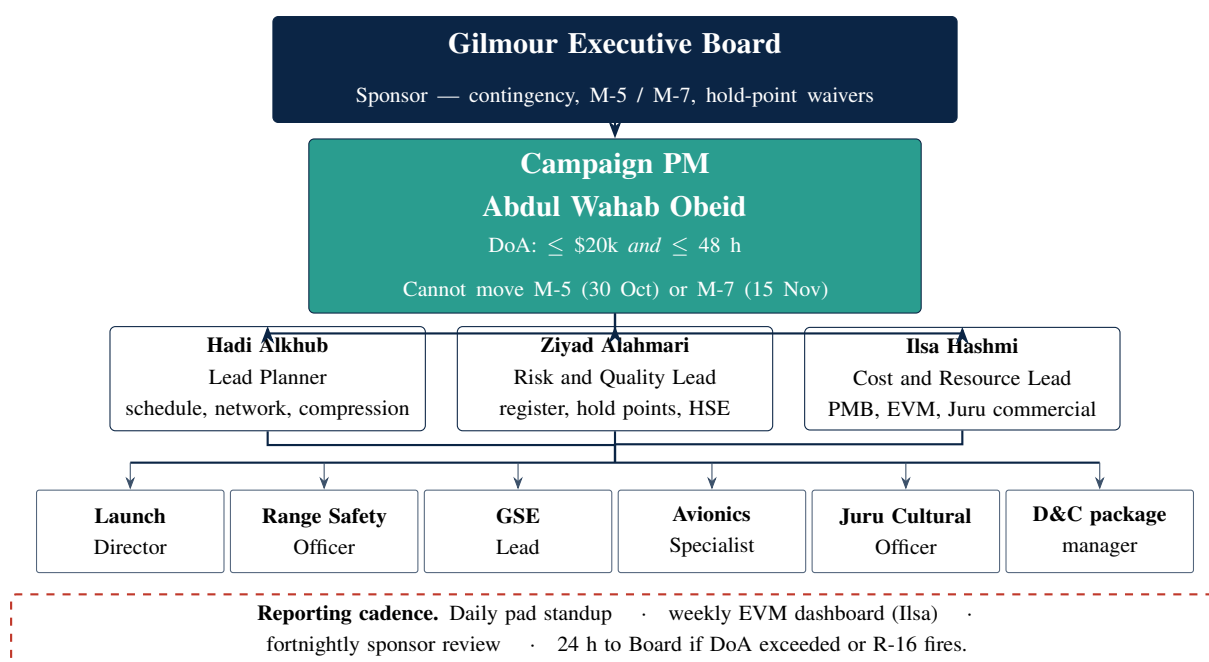


Figure 7.1: Campaign governance for Bowen pad operations. Named roles, a numeric DoA on the PM, and the reporting cadence. Field titles match Table 5.1. The Board, not the PM, moves M-5 or M-7 or draws the \$225,000 reserve.

7.2 Delivery model and procurement

The chosen delivery model is a managing-contractor / design-and-construct hybrid (Turner, 2014; Project Management Institute, 2021). Gilmour Space Technologies retains propulsion intellectual property, vehicle integration and the Australian launch licence. Civil works and cryogenic GSE packages are let as design-and-construct, with an interface-control freeze gate and liquidated damages. That split is the contractual answer to R-07 and R-08. If the D&C GSE interface control drawing lags Gilmour’s propulsion ICD, the lag is an LD event on the contractor, not a Gilmour variation (R-07). If a latent pad-civil defect appears under the hold-down, defect liability remains with the D&C package after HP-1 NDT, which is why R-08 carries EMV \$0 to Gilmour in Table 6.1. Abdul stays Accountable on that row; the package manager stays Responsible.

Turnkey EPC was considered and rejected. After TestFlight 1’s pump failure, handing the propulsion stack and the launch authorisation to a single EPC would put flight-critical intellectual property and statutory accountability in a contractor who does not hold the licence (Gilmour Space Technologies, 2025; Australian Space Agency, 2025). Two tests decide the model (Turner, 2014). Scope certainty is high on site operations—seventeen Level-3 packages, locked dates, a \$1,500,000 work estimate—and low on the vehicle, which remains Gilmour’s; an EPC wants both. Client capability is the other test: Gilmour can integrate a launch vehicle; it should not pour pad grout or run a 100 t crawler as a core competence. Alliance contracting was the second rejected alternative: a four-month campaign cannot carry pain-share overhead, and GBRMPA

and CASA duties are not commercially shareable (Great Barrier Reef Marine Park Authority, 2019; Civil Aviation Safety Authority, 2024). Factory manufacture and freight stay out of scope, so D&C packages stop at the receiving gate and A-100 remains a zero-dollar external wait.

7.3 Quality

Quality is specified against AS9100D for aviation, space and defence and ISO 9001:2015 for the process system (SAE International, 2016; International Organization for Standardization, 2015). Four hold points, each tied to a register row, are the verification architecture; they are listed in Table 7.1. HP-1 is pad tie-down weld NDT on A-123 and is the physical close of R-08. HP-2 is the GSE helium leak test on A-124 and is the physical close of R-05 (inadequate purge before LOX). HP-3 is the pre-static-fire ignition interlock on A-133 and is the physical close of R-01 (residual pump-insulation abort). HP-4 is LRR sign-off on A-141. Work does not proceed past a hold without Ziyad’s release or a Board waiver, and a Board waiver is outside Abdul’s DoA.

Table 7.1: Quality hold points for the Bowen campaign. Each hold is a named artefact, not a meeting. Measurement rules in Section 7.5 follow the same IDs.

Hold	Activity	Risk	Artefact	Release
HP-1	A-123 (15 Oct)	R-08	Pad tie-down weld NDT	D&C mgr / Ziyad
HP-2	A-124 (12 Oct)	R-05	GSE helium leak test	GSE Lead / Ziyad
HP-3	A-133 (30 Oct)	R-01	Pre-static-fire ignition interlock	GSE Lead / Launch Director / Ziyad
HP-4	A-141 (10 Nov)	R-16	LRR sign-off	Launch Director / Abdul / Ziyad

Measurement follows the holds. Discrete tests and permits—A-113, A-132, A-133, A-141, A-143—are 0/100: earned value is zero until the artefact exists. Installations—A-123 and A-124—are milestone percent-complete tied to HP-1 and HP-2. That is why LRR waits on A-134 rather than on A-133: HP-4 cannot be signed on incomplete post-fire data, which is the same AS9100D logic that rejected Option A in Section 3.3 (SAE International, 2016). Fast-tracking LRR over diagnostics would be a hold-point failure, not a schedule saving.

7.4 Health, safety and environment

HSE is this pad under the *Work Health and Safety Act 2011* (Qld), not a generic construction checklist (Queensland Government, 2011). Liquid oxygen is the process hazard: LOX PPE, oxygen-clean tools, and exclusion during load and detank on A-132 and A-133. Blast occupancy is a 1.5 km zone and twelve people—the same cap already drawn on Figure 5.1. Acoustic treatment is a water deluge sized so static-fire level at Bowen township stays at or below 135 dB (R-17), with community notice fourteen days prior. Scrub-fire breaks sit on the pad perimeter because static-fire slag is an ignition source in dry October. Juru artefact buffers and stop-work

authority (R-04) are HSE controls, not a communications plan: A-114 feeds A-132 as a cultural hold, and monitors on the pad can halt WDR. Marine-mammal protocol on A-142 (R-15) is owned by the RSO inside a four-day exclusion window that Option C does not crash (Great Barrier Reef Marine Park Authority, 2019). Fatigue from Option B's 24-hour diagnostic war room is why that crash was rejected; Option C keeps A-142 and the staggered ten-hour shifts already in the resource baseline. Occupancy never exceeds twelve on the mitigated histogram: R-13 is already treated in the \$1,500,000 base by moving A-125 and hiring four surge technicians, not by a later HSE waiver.

7.5 Earned value management

Earned value is the instrument that will tell the Board whether the \$1,725,000 authorisation is still intact (Project Management Institute, 2019; Fleming and Koppelman, 2016; Kerzner, 2017). It is a framework. The campaign has not started, so there is no fourteen-week EV/AC history to plot.

The work performance measurement baseline (distributed planned value) is \$1,500,000, identical to Table 4.1 and to the S-curve that terminates at that figure in Figure 4.1. Contingency of \$225,000 is undistributed budget: it is not December cash and it is not monthly PV. Authorised BAC is \$1,725,000. Option C's \$21,000 is not in the work PMB; if R-12 or R-14 fires and the Board invokes it, the draw appears as actual cost charged to contingency.

SV and CV are currency, never days:

$$SV = EV - PV, \quad CV = EV - AC, \quad SPI = EV/PV, \quad CPI = EV/AC. \quad (5)$$

In-flight estimate at completion on the work baseline is

$$EAC = AC + (BAC_{\text{work}} - EV)/CPI, \quad (6)$$

with $BAC_{\text{work}} = \$1,500,000$. The same CPI is also applied to the authorisation ceiling as BAC/CPI , which is reported against \$1,725,000 so that a green week cannot silently spend the reserve (Project Management Institute, 2019). The finish-line rule is separate from the in-flight bands: final $AC \leq \$1,725,000$. A Green CPI of 0.96 in October is allowed; finishing above BAC is not.

Thresholds that trigger action, used only in this section, are:

- **Green** — $0.95 \leq CPI, SPI \leq 1.05$: weekly dashboard only.
- **Amber** — $0.90 \leq \text{index} < 0.95$: Exception Report and a seven-day recovery plan within 48 hours.
- **Red** — $\text{index} < 0.90$, or $T-0$ slip > 2 days: contingency freeze and Board intervention. SV is not written in days.

The only dashboard numbers permitted in this PEP are the two illustrative readings at M-4 (15 October 2026) in Table 7.2 and Figure 7.2. Planned value at that date is \$825,000: stacking 0/100 complete, GSE HP-2 complete (A-124 finishes before M-4). The on-plan illustration books EV \$810,000 and AC \$840,000, so $CPI = 810,000/840,000 = 0.96$ and $SPI = 810,000/825,000 = 0.98$ —Green, weekly dashboard. The \$15,000 EV–PV gap is 0/100 on remaining A-113 permit hours (permits finish 24 October), not a stacking slip. Even so,

$$EAC = 840,000 + (1,500,000 - 810,000)/0.96 = \$1,558,750,$$

and $BAC/CPI = 1,725,000/0.96 = \$1,796,875$, which sits above the \$1,725,000 ceiling. Green in October still carries a recovery obligation toward the finish-line rule.

The late-stack illustration holds PV at \$825,000 but drops EV to \$760,000 and AC to \$830,000 because HP-1 is unsigned. Then $CPI = SPI = 0.92$ —Amber. Ilsa issues the Exception Report within 48 hours and a seven-day recovery that either completes HP-1 or takes a Board decision. Work EAC is $\$830,000 + (1,500,000 - 760,000)/0.92 = \$1,634,348$. Those two points are a worked example at one physical gate, not a time series and not a claim that October has already been worked.

Table 7.2: Illustrative earned-value status at M-4 (15 October 2026). The campaign has not been executed. PV is the locked work-PMB value at stacking complete. Indices use the reported two-decimal CPI/SPI from EV/AC and EV/PV .

Status at M-4	PV	EV	AC	CPI	SPI	Action
Planned (stacking 0/100 complete, GSE HP-2 complete)	825,000	—	—	—	—	—
Illustrative on-plan	825,000	810,000	840,000	0.96	0.98	Green — weekly dashboard
Illustrative late stack	825,000	760,000	830,000	0.92	0.92	Amber — 48 h exception report

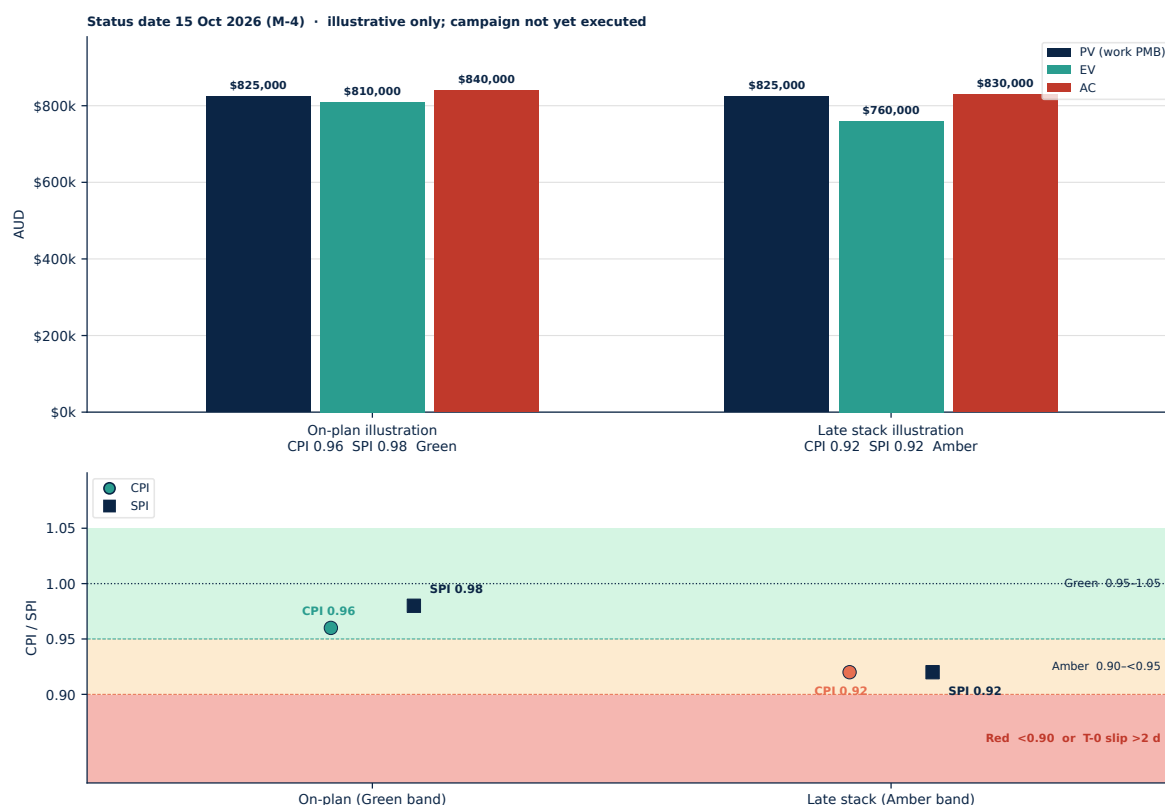


Figure 7.2: Illustrative status at M-4; campaign not yet executed. Upper panel: PV, EV and AC at the single status date 15 October 2026 for the on-plan and late-stack cases. Lower panel: Green / Amber / Red CPI–SPI bands, with the on-plan pair (0.96, 0.98) in Green and the late-stack pair (0.92, 0.92) in Amber. Red is also triggered by $T-0$ slip greater than 2 days. No weekly EV/AC history is implied.

8 Integration and professionalism

This PEP is one campaign described eight times. Scope, WBS, network, dollars, occupancy, register, hold points and earned-value bands name the same Bowen pad, the same seventeen cost accounts, and the same 15 November $T-0$. Where a view disagrees with Table 8.1, `data/pep_baseline.yaml` is the book of record.

8.1 Document consistency

The identities are already in the body. The work estimate is \$1,500,000 in Objective 2, Table 4.1, the last row of Table 4.3, the end of Figure 4.1, and the distributed PMB in Section 7.5. Contingency is \$225,000 because eleven cost-bearing rows in Table 6.1 sum to \$175,500 of EMV and an in-scope RES allowance of \$49,500 is added; 15.00% is that result. Authorised BAC is \$1,725,000, and $AC \leq BAC$ is the same finish-line in Objective 2 and in the earned-value framework. Dates are activity early finishes: M-4 is A-123 on 15 October (also the only earned-value status date); M-5 is A-133 on 30 October after cold WDR; M-6 is A-141 after diagnostics;

M-7 is A-143 on 15 November; M-8 is A-144 on 20 November. Option C's 10 November lift-off is a contingent \$21,000 draw if R-12 or R-14 fires; it is not in Table 4.3 and is not redrawn on Figure 3.2. Occupancy peaks at twelve in Figure 5.1 after A-125 is parked on float and \$22,400 of surge labour is placed inside WBS 1.3.3—the same cap as R-13 and Section 7.4.

Table 8.1 is that thread in one view. The previous-draft activity series that mixed RF calibration with cryogenic GSE is retired and is not used. *SV* is currency, never days. The S-curve is not forced to \$1,725,000. Eight people are the organic crew, not a hard pad limit.

Table 8.1: Golden-thread reconciliation. Every view in this PEP is required to show the value in the middle column; the right-hand column is where the marker should look.

Thread	Locked value	Views that must agree
Work PMB	\$1,500,000	Sec. 2.4; Table 4.1; Table 4.3 (ends here); Fig. 4.1; Sec. 7.5 distributed <i>PV</i>
Contingency	\$225,000 = EMV \$175,500 + RES \$49,500	Table 4.2; Table 6.1 dollar column; 15.00% is the <i>result</i>
Authorised <i>BAC</i>	\$1,725,000	Finish-line $AC \leq BAC$ in Sec. 2.4 and Sec. 7.5
M-4 stacking	15 Oct 2026 = A-123 <i>EF</i>	Table 2.1; Table 3.1; Fig. 3.2; Fig. 7.2 status date
M-5 static fire	30 Oct 2026 = A-133 <i>EF</i>	Cold WDR A-132 then hot fire A-133; HP-3 on A-133
M-6 LRR	10 Nov 2026 = A-141 <i>EF</i>	LRR waits on A-134, not on A-133; HP-4
M-7 <i>T-0</i>	15 Nov 2026 = A-143 <i>EF</i>	Baseline. Option C 10 Nov 2026 is contingent only; not redrawn on Fig. 3.2
M-8 handover	20 Nov 2026 = A-144 <i>EF</i>	Five days after M-7; Table 2.1 closes the 22 Nov collision
Pad occupancy	organic 8 / cap 12 / unmit. 14 / mit. 12	Fig. 5.1; R-13; Sec. 7.4
Surge labour	\$22,400 inside WBS 1.3.3	In the \$1,500,000 base, not a contingency draw
Option C	\$21,000 from contingency if R-12/R-14	Not in Table 4.3; Board draw under the Sec. 7.1 DoA
Contribution	$25.0\% \times 4 = 100.0\%$ (168 h equalised)	Title page; Table A.1

8.2 Team fallback and contribution

Team-member fallback is a three-tier operating procedure on the work, not a single redistribution bullet. If a member is non-responsive for more than 48 hours: (1) the Lead Planner (Hadi Alkhub) audits the artefacts within 12 hours against the WBS package in Table 5.1; (2) the Campaign PM (Abdul Wahab Obeid) issues a written addendum reassigning those packages; (3) the Buddycheck Personal Assessment Form is locked against Microsoft Teams and version

history so contribution evidence cannot be rewritten after the fact. The procedure reallocates rows in Table 5.1. It does not move $T-0$, BAC , or a hold point. Pad resource conflicts still use the 4 h / 12 h / 24 h SLA in Section 5.3; contingency draws and M-5 / M-7 moves remain Board decisions under Section 7.1.

Contribution is equalised at 42 hours per member so the cover split of 25.0% each remains 100.0%. Table A.1 and Appendix A are that split: Hadi Alkhub on scope and schedule; Ilsa Hashmi on cost, resources and EVM; Ziyad Alahmari on the register, quality and HSE; Abdul Wahab Obeid on the executive summary, integration and final QA. Hours, not seniority, set the percentage.

8.3 Presentation and referencing

The document is 12 point Times on A4, 25 mm margins, 1.5 line spacing, numbered headings to two levels. Figures and tables are numbered within the section, captioned, and referred to in the prose before the float. TikZ draws node diagrams (WBS, governance). Matplotlib reads the YAML register for data plots (Gantt, compression, S-curve, histogram, risk matrix, earned-value example). Citations are APA 7th author–year through `natbib`. Design choices are cited where they are made: critical path and compression (Project Management Institute, 2021; Kerzner, 2017); hold points (SAE International, 2016); contingency as expected monetary value (Kerzner, 2017; International Organization for Standardization, 2018); labour and plant (Fair Work Ombudsman, 2020; Rawlinsons, 2025); range permissions (Civil Aviation Safety Authority, 2024; Great Barrier Reef Marine Park Authority, 2019); delivery model (Turner, 2014); earned value (Project Management Institute, 2019; Fleming and Koppelman, 2016); this pad’s purpose (Gilmour Space Technologies, 2025; Australian Space Agency, 2025; Dougherty, 2026). Word count on the title page excludes references, appendices, and content inside tables and figures.

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A Contribution appendix

Table A.1: Agreed contribution after hours equalisation. Each member 42 h; percentages match the title page and sum to 100.0%.

Team member	Hours	Primary deliverables	%
Abdul Wahab Obeid (s5303839)	42	Executive summary; integration; governance; delivery model; final QA	25.0
Hadi Alkhub (s5331680)	42	Scope; schedule; network logic; Gantt; compression	25.0
Ziyad Alahmari (s5375826)	42	Risk register; quality and HSE	25.0
Ilsa Hashmi (s5310047)	42	Cost; resources; capacity; EVM framework	25.0
Total	168		100.0

Hours are equalised so that the 25.0% shares on the title page remain valid. The three-tier fallback in Section 8.2 (12-hour artefact audit, PM WBS addendum, Buddycheck PAF lock) is the procedure that would reallocate these rows.

We confirm that the contribution percentages above are agreed by all members.

Abdul Wahab Obeid

Signature: _____

Date: _____

Ziyad Alahmari

Signature: _____

Date: _____

Hadi Alkhub

Signature: _____

Date: _____

Ilsa Hashmi

Signature: _____

Date: _____